

Section : Physics

- Q.1** According to the Biot- Savart law, the magnetic field due to a current element is
(A) inversely proportional to the square of the distance from the current element to the point of interest.
(B) produced by a vector source.
(C) perpendicular to the plane containing the displacement vector and the current element.
(D) non-zero at any point in the direction of the current element.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

- Q.2** The phenomenon responsible for the blue colour of the sky

1. Reflection of light
2. Refraction of light
3. Scattering of light
4. Dispersion of light

Options

1. 1
2. 2
3. 3
4. 4

- Q.3** The energy required to remove an electron from the inner most orbit of the hydrogen atom to a distance far away from its nucleus is

1. -13.6 eV
2. 13.6 eV
3. 3.4 eV
4. 0 eV

Options

1. 1
2. 2
3. 3
4. 4

- Q.4** According to Rayleigh's law, the intensity of scattered light varies with wavelength λ as

1. $\propto \lambda^4$
2. $\propto \frac{1}{\lambda^4}$
3. $\propto \lambda^2$
4. $\propto \frac{1}{\lambda^2}$

Options

1. 1
2. 2
3. 3
4. 4

- Q.5** To form n-type semiconductor from a pure semiconductor, the impurity to be added is

1. Aluminium
2. Boron
3. Indium
4. Arsenic

Options

1. 1
2. 2
3. 3
4. 4

- Q.6** A radioactive nucleus of mass number A and atomic number Z, emits an alpha particle first and then a beta negative particle. The mass number and atomic number of the daughter nucleus will be

1. A-4, Z-1
2. A-1, Z-2
3. A-2, Z + 1
4. A-2, z + 2

Options

1. 1
2. 2
3. 3

- Q.7** A particle of charge q and mass m moves in a circular orbit of radius r with angular speed ω . The ratio of the magnitude of its magnetic moment (m) to that of its angular momentum (L) depends on
1. m and ω only
 2. m and q only
 3. m , ω and q
 4. ω and q only

Options

1. 1
2. 2
3. 3
4. 4

- Q.8** The electric potential due to an electric dipole depends on
1. distance from the dipole only
 2. dipole moment of the dipole only
 3. dipole moment and distance from the dipole only
 4. dipole moment, distance from the dipole and the angle between the position vector and the dipole moment vector

Options

1. 1
2. 2
3. 3
4. 4

- Q.9** A capacitor has a reactance of $100\ \Omega$ at 50 Hz . Its reactance at 125 Hz would be
1. $50\ \Omega$
 2. $30\ \Omega$
 3. $40\ \Omega$
 4. $125\ \Omega$

Options

1. 1
2. 2
3. 3
4. 4

- Q.10** Self-inductance of a solenoid is
1. directly proportional to the current flowing through the coil
 2. directly proportional to the number of turns per unit length
 3. directly proportional to its area of cross-section
 4. inversely proportional to its area of cross-section

Options

1. 1
2. 2
3. 3
4. 4

- Q.11** Which of the following statements are true for a conductor in an external electric field?
- (A) Inside a conductor, the electric field is non-zero.
- (B) At the surface of a charged conductor, the electric field must be normal to the surface at every point.
- (C) The interior of a conductor can have no excess charge in a static situation.
- (D) Electrostatic potential is constant throughout the volume of the conductor and has the same value on its surface.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

- Q.12** As the mass number A increases, which of the following quantities related to a nucleus does not change?
1. Mass
 2. Volume
 3. Density
 4. Binding Energy

Options

1. 1
2. 2
3. 3
4. 4

- Q.13** A transmitting station transmits radio waves of wavelength 300 m . The inductance of the coil required with a capacitor of capacity $1.0\ \mu\text{F}$ in a resonant circuit will be

1. $1.27 \times 10^{-9} \text{ H}$
2. $2.53 \times 10^{-8} \text{ H}$
3. $2.73 \times 10^{-6} \text{ H}$
4. $2.53 \times 10^{-7} \text{ H}$

Options

1. 1
2. 2
3. 3
4. 4

Q.14 By increasing the temperature, the specific resistance of a conductor and a semiconductor

1. increases for both
2. decreases for both
3. increases and decreases, respectively
4. decreases and increases, respectively

Options

1. 1
2. 2
3. 3
4. 4

Q.15 Match List-I with List-II

List-I

(Quantity)

(A) Einstein's Photoelectric Equation

(B) De-Broglie Wavelength

(C) Stopping potential

(D) Threshold wavelength

List-II

(Formula)

(I) K_{max}/e

(II) $K_{max} = hc/\lambda - \phi$

(III) hc/ϕ

(IV) h/p

Choose the correct answer from the options given below:

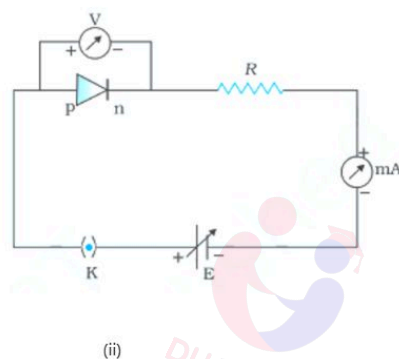
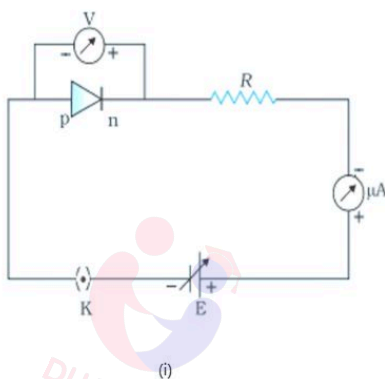
1. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
2. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.16

The experimental circuit arrangements for studying V-I characteristics of a p-n junction diode are given below:



Identify the circuit for forward bias and for reverse bias.

1. Circuit (i) forward bias circuit (ii) reverse bias
2. Circuit (i) reverse bias circuit (ii) forward bias
3. circuit (i) forward bias circuit (ii) forward bias
4. Circuit (i) reverse bias circuit (ii) reverse bias

Options

1. 1
2. 2
3. 3
4. 4

Q.17 A charge of $50\ \mu\text{C}$ is uniformly distributed on the surface of a thin spherical shell of radius 10 cm. The electric field at a point at a distance of 5 cm from the center of the shell is

1. $18 \times 10^7\ \text{N/C}$
2. $9 \times 10^7\ \text{N/C}$
3. $9 \times 10^6\ \text{N/C}$
4. Zero

Options

- 1.
- 2.
- 3.
- 4.

Q.18 An ammeter of resistance $5\ \Omega$ reads up to 1 A. The value of the shunt required to increase its range to 5 A will be

1. $12.5\ \Omega$
2. $1.25\ \Omega$
3. $1.00\ \Omega$
4. $0.125\ \Omega$

Options

- 1.
- 2.
- 3.
- 4.

Q.19 The magnetic field at a distance r from a long wire carrying current I is 0.4 T. The magnetic field at a distance $2r$ from the wire is

1. 0.2 T
2. 0.8 T
3. 0.1 T
4. 1.6 T

Options

- 1.
- 2.
- 3.
- 4.

Q.20 An electron is moving at a velocity of 100 m/s along the X-axis in the presence of an electric field of 30 V/m directed along the Y-axis and a magnetic field of 300 G directed along the Z-axis. The magnitude and direction of force acting on the electron is
(Given: the charge on the electron is " e ")

1. $33\ e\ \text{N}$ along the Y-axis
2. $30\ e\ \text{N}$ along the negative Y-axis
3. $27\ e\ \text{N}$ along the Y-axis
4. $27\ e\ \text{N}$ along the negative Y-axis

Options

- 1.
- 2.
- 3.
- 4.

Q.21 A metal cube of side 0.1 m is uniformly charged by $18\ \mu\text{C}$. Its surface charge density will be

1. $18.8 \times 10^{-4}\ \text{C m}^{-2}$
2. $18 \times 10^{-3}\ \text{C m}^{-2}$
3. $3 \times 10^{-4}\ \text{C m}^{-2}$
4. $3 \times 10^{-3}\ \text{C m}^{-2}$

Options

- 1.
- 2.
- 3.
- 4.

Q.22 The meter bridge is a device

1. to compare potential differences
2. to measure potential difference
3. to measure resistance
4. to compare emfs of two sources

Options

- 1.
- 2.
- 3.
- 4.

Q.23 A light with an energy flux of $25\ \text{W/cm}^2$ falls on a non-reflecting surface at normal incidence. If the surface has an area of $50\ \text{cm}^2$ and the total energy falling on the surface is $10 \times 10^5\ \text{J}$. The total duration of time of the incidence of light is

1. 200 s
2. 400 s
3. 600 s
4. 800 s

Options

1. 1
2. 2
3. 3
4. 4

Q.24 The ratio of the electric field (E) and the magnetic field (B) associated with a plane electromagnetic wave is equal to

1. ωk
2. ωk^{-1}
3. ω^{-1}
4. ω

Options

1. 1
2. 2
3. 3
4. 4

Q.25 The following experiment suggests that light propagates as waves

1. Focault's method of finding the velocity of light in water
2. Einstein photoelectric emission experiment
3. Young's Double slit experiment
4. Compton's experiment on scattering of light

Options

1. 1
2. 2
3. 3
4. 4

Q.26 The primary voltage and primary current of a transformer are 220 V and 10 A. The ratio of the number of turns on the secondary coil to the number of turns on the primary coil is 2:1. The values of voltage and current at the output are

1. 110 V and 20 A
2. 110 V and 5 A
3. 440 V and 5 A
4. 440 V and 20 A

Options

1. 1
2. 2
3. 3
4. 4

Q.27 If we double the number of turns in a moving coil galvanometer, the voltage sensitivity of the galvanometer

1. Increases by 2 times
2. Increases by 3 times
3. increases by 4 times
4. does not change

Options

1. 1
2. 2
3. 3
4. 4

Q.28 A change of current from 6.0 A to 3.0 A in 300 ms in a coil produces an induced emf of 0.5 V in the neighboring coil. The mutual inductance between the two coils would be

1. 0.5 H
2. 0.05 H
3. 0.03 H
4. 0.2 H

Options

1. 1
2. 2
3. 3
4. 4

Q.29 Identify the correct statement(s) among the following:

- (A) An electric bulb connected with a capacitor in series with electric DC source, glows with full brightness.
- (B) An electric bulb connected with capacitor in series with an AC source, does not glow.
- (C) An electric bulb connected with capacitor in series with a electric DC source, does not glow.
- (D) An electric bulb connected with capacitor in series with an AC source, glows well.

Choose the correct answer from the options given below:

1. (A) only
2. (A) and (B)

3. (B) and (C)
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.30 An astronomical telescope uses two lenses of power 10 D and 2 D. The distance between the objective and the eye lens in normal adjustment would be

1. 20 cm
2. 30 cm
3. 60 cm
4. 8.3 cm

Options 1. 1

2. 2
3. 3
4. 4

Q.31 In a diffraction pattern due to a narrow slit of width a , alternate dark and bright fringe pattern is obtained on a screen kept at distance D from the slit. Choose the correct option.

- (A) The angular width of central maximum is given by $2D/a$, linear width is given by $2D\lambda/a$
(B) When the size of aperture, a is quite large, the diffraction effect is negligible,
(C) For light of smaller wavelength, angular width of central maximum increases, while linear width decreases,
(D) Angular width of central maximum does not decrease while linear width decreases on decreasing distance of screen from slit.

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A) and (C) only
3. (B) and (C) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Two batteries of emf 5 V with internal resistance $2\ \Omega$ and 6 V with internal resistance $3\ \Omega$ are connected in series with an external resistor. If the current in the circuit is 0.5 A, then the resistance of the resistor is

1. 15 Ω
2. 17 Ω
3. 19 Ω
4. 21 Ω

Options 1. 1

2. 2
3. 3
4. 4

Q.33 In an electrical circuit, the three resistors of $3\ \Omega$, $6\ \Omega$ and $9\ \Omega$ are connected in series. The voltage drop across $3\ \Omega$ is 5

V. The voltage drop across the combination is

1. 5 V
2. 30 V
3. 9 V
4. 15 V

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Match List-I with List-II

List-I

(Quantity)

(A) Electric field due to a point charge

(B) Electric potential due to a point charge

(C) Electric field due to uniformly charged infinite plane sheet

(D) Electric field due to an electric dipole

List-II

(Variation with distance)

(I) Inversely proportional to the cube of the distance

(II) Inversely proportional to the square of the distance

(III) Inversely proportional to the distance

(IV) Independent of the distance

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
2. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.35 The internal resistance of a cell of emf 2 V is 0.1 Ω . It is connected to an external resistance of 3.9 Ω . The terminal potential difference across the cell will be

1. 0.5 V
2. 1.5 V
3. 1.95 V
4. 2 V

Options

1. 1
2. 2
3. 3
4. 4

Q.36 Two bulbs marked 220 V, 200 W and 220 V, 1000 W, respectively, are connected in series to 220 V mains supply. The ratio of heat generated in them, respectively will be

1. 1:5
2. 5:1
3. 1:25
4. 25:1

Options

1. 1
2. 2
3. 3
4. 4

Q.37 Arrange the following in the increasing order of their energy band gap

- (A) Si
- (B) Ge
- (C) C
- (D) Sn

Choose the correct answer from the options given below:

1. (B), (A), (C), (D)
2. (D), (B), (A), (C)
3. (B), (A), (D), (C)
4. (C), (B), (A), (D)

Options

1. 1
2. 2
3. 3
4. 4

Q.38 Two parallel conductors each carrying a current of 10 mA in opposite directions are placed at a distance 10 cm apart.
The force per unit length between two conductors is

1. $10 \times 10^{-4} \left(\frac{\mu_0}{2\pi} \right) \text{ N/m}$
2. $5 \times 10^{-4} \left(\frac{\mu_0}{2\pi} \right) \text{ N/m}$
3. $20 \times 10^{-4} \left(\frac{\mu_0}{2\pi} \right) \text{ N/m}$
4. $1 \times 10^{-4} \left(\frac{\mu_0}{2\pi} \right) \text{ N/m}$

Options

1. 1
2. 2
3. 3
4. 4

Q.39 To demonstrate the phenomenon of interference, required two sources of light must have

1. nearly the same frequency only
2. different wavelengths
3. same frequency and have a constant phase difference
4. same frequency and have a rapidly changing phase difference

Options

1. 1
2. 2
3. 3
4. 4

Q.40 A ray of light is incident on an equilateral prism at an angle of 45° and is refracted parallel to the base of the prism.

The angle of deviation would be

1. 45°
2. 30°
3. 60°
4. 50°

Options

1. 1
2. 2
3. 3
4. 4

Q.41 A parallel beam of monochromatic light of wavelength 489 nm passes through a single slit of width 0.3 mm.

The angular width of the central maxima is

1. $1.63 \times 10^{-6} \text{ rad}$
2. $6.13 \times 10^2 \text{ rad}$
3. $1.63 \times 10^{-3} \text{ rad}$
4. $3.26 \times 10^{-3} \text{ rad}$

Options

1. 1
2. 2
3. 3
4. 4

Q.42 Two point charges of $1 \times 10^{-6} \text{ C}$ and $4 \times 10^{-6} \text{ C}$ are placed at a certain distance away from each other. The resultant electric field intensity is zero at a distance of 15 cm from the charge $1 \times 10^{-6} \text{ C}$. The distance between the two point charges will be :

1. 15 cm
2. 30 cm
3. 45 cm
4. 5 cm

Options

1. 1
2. 2
3. 3
4. 4

Q.43 A battery of 200 V charges a capacitor to $10 \times 10^{-6} \text{ C}$. Now the battery is disconnected, and the plate separation is doubled.
The potential of the capacitor will be

1. 400 V
2. 200 V
3. 100 V
4. 0 V

Options 1. 1

2. 2

3. 3

4. 4

Q.44 If the wavelength of an electromagnetic radiation is doubled, the energy of the photons would be

1. double of its initial value
2. unchanged
3. one half of its initial value
4. four times of its initial value

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Which of the following statements are correct?

- (A) Metals obey Ohms law at constant temperature.
- (B) The resistivity of the metals depends on temperature.
- (C) The conductivity of the metals is independent of temperature.
- (D) The number of electrons per unit volume in a metal is independent of temperature.

Choose the *correct* answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Arrange the wavelengths of the following radiations in the decreasing order:

λ_{visible} , $\lambda_{X\text{-rays}}$, $\lambda_{\text{radiowaves}}$, $\lambda_{\gamma\text{-rays}}$

1. $\lambda_{\text{visible}} > \lambda_{X\text{-rays}} > \lambda_{\text{radiowaves}} > \lambda_{\gamma\text{-rays}}$
2. $\lambda_{\text{visible}} > \lambda_{X\text{-rays}} > \lambda_{\gamma\text{-rays}} > \lambda_{\text{radiowaves}}$
3. $\lambda_{X\text{-rays}} > \lambda_{\text{radiowaves}} > \lambda_{\gamma\text{-rays}} > \lambda_{\text{visible}}$
4. $\lambda_{\text{radiowaves}} > \lambda_{\text{visible}} > \lambda_{X\text{-rays}} > \lambda_{\gamma\text{-rays}}$

Options 1. 1

2. 2

3. 3

4. 4

Q.47 In a series RLC circuit at resonance

- (A) Impedance is maximum
- (B) $X_c = X_L$
- (C) $Z = R$

(D) $\omega = \frac{1}{\sqrt{LC}}$

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.48 In the atomic hydrogen line spectrum, the wavelength of the first line of the Lyman series is

(Given: Rydberg constant $R = 1.097 \times 10^7 m^{-1}$)

1. 91 nm
2. 122 nm
3. 365 nm
4. 656 nm

Options 1. 1

2. 2

3. 3

4. 4

Q.49 A bullet of mass 40 g is traveling at a speed of 1.0 km/s. Its de-Broglie wavelength would be

1. 1.7×10^{-35} m

2. 1.1×10^{-32} m

3. 3.0×10^{-25} m

4. 2.8×10^{-32} m

Options

1. 1

2. 2

3. 3

4. 4

Q.50 For a parallel plate capacitor having plate area $1.13 \times 10^3 \text{ m}^2$ and separation between plates 0.5 cm, the capacitance is

1. 2.8 μF

2. 2.0 μF

3. 2.0 nF

4. 4.0 μF

Options

1. 1

2. 2

3. 3

4. 4